Supplementary Material: Analyzing Categorical Time Series in the Presence of Missing Observations

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S Supplement

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S.1 Derivations of Section 3.2

The covariances in (3.5) satisfy

$$Cov\big[Z^*_{t,i},\ Z^*_{s,j}\big] \;=\; \begin{cases} Cov[O_t,\ O_s] \;= \gamma_O(|t-s|) & \text{if } i=j=-1, \\ Cov[O_t,\ O_s\, Z_{s,j}] & \text{if } i=-1<j, \\ Cov[O_t\, Z_{t,i},\ O_s\, Z_{s,j}] & \text{if } i,j>-1. \end{cases}$$

The formula for $i=j=-1$ allows to conclude on $\sigma_{-1,-1}$ in (3.6). By the independence of (O_t) to (X_t), we have

$$\begin{aligned} Cov[O_t,\ O_s\, Z_{s,j}] &= E[O_t\, O_s\, Z_{s,j}] - E[O_t]\, E[O_s\, Z_{s,j}] \\ &= E[O_t\, O_s]\, E[Z_{s,j}] - E[O_t]\, E[O_s]\, E[Z_{s,j}] \\ &= Cov[O_t, O_s]\, E[Z_{s,j}] \;=\; \gamma_O(|t-s|)\, f_j, \end{aligned}$$

so $\sigma_{i,-1} = \sigma_{-1,i} = f_i\, \sigma_{-1,-1}$ for $i>-1$. Analogously, we get

$$\begin{aligned} Cov[O_t\, Z_{t,i},\ O_s\, Z_{s,j}] &= E[O_t\, O_s]\, E[Z_{t,i}\, Z_{s,j}] - E[O_t]\, E[O_s]\, E[Z_{t,i}]\, E[Z_{s,j}] \\ &= E[O_t\, O_s]\, Cov[Z_{t,i}, Z_{s,j}] + Cov[O_t, O_s]\, E[Z_{t,i}]\, E[Z_{s,j}] \\ &= \pi(|t-s|)\, Cov[Z_{t,i}, Z_{s,j}] + \gamma_O(|t-s|)\, f_i f_j \end{aligned}$$

for $i,j>-1$. Note that

$$Cov[Z_{t,i}, Z_{s,j}] \;=\; \begin{cases} f_{ij}(t-s) - f_i f_j & \text{if } t\geq s, \\ f_{ji}(s-t) - f_i f_j & \text{if } s\geq t, \end{cases} \qquad \text{with } f_{ij}(0) = f_{\min\{i,j\}}.$$

So for $i,j>-1$, we get

$$\begin{aligned} \sigma_{i,j} &= Cov[O_0\, Z_{0,i},\ O_0\, Z_{0,j}] \;+\; \textstyle\sum_{h=1}^{\infty} \Big(Cov[O_0\, Z_{0,i},\ O_h\, Z_{h,j}] + Cov[O_h\, Z_{h,i},\ O_0\, Z_{0,j}] \Big) \\ &= f_i f_j\, \sigma_{-1,-1} \;+\; \pi\,\big(f_{\min\{i,j\}} - f_i f_j\big) \;+\; \textstyle\sum_{h=1}^{\infty} \pi(h)\, \Big(f_{ij}(h) + f_{ji}(h) - 2\, f_i f_j \Big). \end{aligned}$$

This completes the proof of (3.6).

Next, we consider the function $\boldsymbol{g}: [0;1]^{m+1} \to [0;1]^m$ with $g_j(x_{-1}, x_0, \ldots, x_{m-1}) = x_j/x_{-1}$ for $j=0,\ldots,m-1$. So $\hat{\boldsymbol{f}}^* = \boldsymbol{g}\big(\overline{\boldsymbol{Z}^*_t}\big)$ and $\boldsymbol{f} = \boldsymbol{g}(\boldsymbol{\mu}_{\boldsymbol{Z}^*})$. The gradient of $g_j(\boldsymbol{x})$ equals $\big(\frac{-x_j}{x_{-1}^2}, 0, \ldots, \frac{1}{x_{-1}}, \ldots, 0\big)$, with $\frac{1}{x_{-1}}$ at position j. Thus, the Jacobian of $\boldsymbol{g}$, evaluated in $\boldsymbol{\mu}_{\boldsymbol{Z}^*} = \pi\,\big(1, \boldsymbol{f}^\top\big)^\top$, equals

$$\mathbf{D} \;=\; \begin{pmatrix} -\frac{f_0}{\pi} & \frac{1}{\pi} & \cdots & 0 \\ \vdots & & \ddots & \\ -\frac{f_{m-1}}{\pi} & 0 & \cdots & \frac{1}{\pi} \end{pmatrix}.$$

Now, we consider the linear Taylor approximation $\hat{\boldsymbol{f}}^* \approx \boldsymbol{f} + \mathbf{D}\,\big(\overline{\boldsymbol{Z}}_t^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\big)$. (3.7) is derived by applying the Delta method to the CLT (3.5), leading to

$$\sqrt{n}\,\Big(\hat{\boldsymbol{f}}^* - \boldsymbol{f}\Big) \ \xrightarrow{\mathrm{d}}\ \mathrm{N}\big(\mathbf{0},\ \boldsymbol{\Sigma}^*\big), \quad \text{where } \boldsymbol{\Sigma}^* = \mathbf{D}\,\boldsymbol{\Sigma}\,\mathbf{D}^\top.$$

Here, the entries σ_{ij}^* with $i,j = 0,\ldots,m-1$ compute as

$$\begin{aligned}
\sigma_{ij}^* \ &= \ \textstyle\sum_{k,l=-1}^{m-1} d_{ik}\, d_{jl}\, \sigma_{kl} \\
&= \ d_{i,-1}\, d_{j,-1}\, \sigma_{-1,-1} \ + \ d_{i,-1}\, d_{j,j}\, \sigma_{-1,j} \ + \ d_{i,i}\, d_{j,-1}\, \sigma_{i,-1} \ + \ d_{i,i}\, d_{j,j}\, \sigma_{i,j} \\
&= \ \tfrac{1}{\pi^2}\,\Big(f_i f_j\, \sigma_{-1,-1} - f_i\, \sigma_{-1,j} - f_j\, \sigma_{i,-1} + \sigma_{i,j}\Big) \ = \ \tfrac{1}{\pi^2}\,\big(\sigma_{i,j} - f_i f_j\, \sigma_{-1,-1}\big).
\end{aligned}$$

In the last step, we used that $\sigma_{i,-1} = \sigma_{-1,i} = f_i\, \sigma_{-1,-1}$ according to (3.6). Using again (3.6), case $i,j > -1$, we get the expression provided by (3.7).

Finally, we consider the quadratic Taylor approximation $\hat{f}_j^* \approx f_j + \mathbf{D}^{(j)}\,\big(\overline{\boldsymbol{Z}}_t^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\big) + \frac{1}{2}\,\big(\overline{\boldsymbol{Z}}_t^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\big)^\top\,\mathbf{H}^{(j)}\,\big(\overline{\boldsymbol{Z}}_t^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\big)$, where $\mathbf{H}^{(j)}$ is the Hessian of g_j evaluated in $\boldsymbol{\mu}_{\boldsymbol{Z}^*}$. Here, the non-zero second-order partial derivatives of g_j are $\frac{\partial^2}{\partial x_{-1}^2}\, g_j = 2x_j/x_{-1}^3$ and $\frac{\partial^2}{\partial x_{-1}\,\partial x_j}\, g_j = -1/x_{-1}^2$, whereas $\frac{\partial^2}{\partial x_j^2}\, g_j = 0$. Evaluated in $\boldsymbol{\mu}_{\boldsymbol{Z}^*}$, this gives the non-zero entries $h_{-1,-1}^{(j)} = \frac{2\,f_j}{\pi^2}$ and $h_{-1,j}^{(j)} = \frac{-1}{\pi^2}$ of the Hessian $\mathbf{H}^{(j)}$. This implies the bias approximation provided by (3.8):

$$n\,E\big[\hat{f}_j^* - f_j\big] \ \approx \ \tfrac{1}{2}\, h_{-1,-1}^{(j)}\, \sigma_{-1,-1} + h_{-1,j}^{(j)}\, \sigma_{-1,j} \ = \ \tfrac{1}{\pi^2}\,\big(f_j\, \sigma_{-1,-1} - \sigma_{-1,j}\big) \ = \ 0,$$

where the last equality follows from (3.6).

S.2 Derivations of Section 3.3

To derive the asymptotics of $\widehat{\text{IOV}}$ from (3.5) and (3.6), define the function

$$g(x_{-1}, x_0, \ldots, x_{m-1}) \;=\; \tfrac{4}{m} \sum_{i=0}^{m-1} \tfrac{x_i}{x_{-1}} \,(1 - \tfrac{x_i}{x_{-1}}),$$

which has the partial derivatives

$$\begin{aligned}
\tfrac{\partial}{\partial x_{-1}}\, g \;&=\; \tfrac{4}{m} \sum_{i=0}^{m-1} \big(-\tfrac{x_i}{x_{-1}^2} + \tfrac{2\,x_i^2}{x_{-1}^3}\big), \qquad & \tfrac{\partial}{\partial x_j}\, g \;&=\; \tfrac{4}{m} \big(\tfrac{1}{x_{-1}} - \tfrac{2\,x_j}{x_{-1}^2}\big);\\
\tfrac{\partial^2}{\partial x_{-1}^2}\, g \;&=\; \tfrac{4}{m} \sum_{i=0}^{m-1} \big(\tfrac{2\,x_i}{x_{-1}^3} - \tfrac{6\,x_i^2}{x_{-1}^4}\big), \qquad & \tfrac{\partial^2}{\partial x_{-1}\,\partial x_j}\, g \;&=\; \tfrac{4}{m} \big(\tfrac{-1}{x_{-1}^2} + \tfrac{4\,x_j}{x_{-1}^3}\big),\\
\tfrac{\partial^2}{\partial x_j\,\partial x_k}\, g \;&=\; \delta_{j,k}\, \tfrac{4}{m} \big(-\tfrac{2}{x_{-1}^2}\big),
\end{aligned}$$

for $j, k = 0, \ldots, m-1$. Evaluated in $\boldsymbol{\mu}_{\boldsymbol{Z}^*} = \pi\,\big(1, \boldsymbol{f}^\top\big)^\top$, we get the Jacobian $\mathbf{D}$ and the Hessian $\mathbf{H}$ defined by the entries

$$\begin{aligned}
d_j \;&=\; \tfrac{4}{m}\,\tfrac{1}{\pi}\,(1 - 2\,f_j), \qquad d_{-1} \;=\; -\tfrac{4}{m}\,\tfrac{1}{\pi} \sum_{i=0}^{m-1} f_i(1 - 2\,f_i) \;=\; -\sum_{i=0}^{m-1} f_i\, d_i;\\
h_{-1,-1} \;&=\; 2\,\tfrac{4}{m}\,\tfrac{1}{\pi^2} \sum_{i=0}^{m-1} f_i(1 - 3\,f_i), \qquad h_{-1,j} \;=\; -\tfrac{4}{m}\,\tfrac{1}{\pi^2}\,(1 - 4\,f_j),\\
h_{j,k} \;&=\; -2\,\delta_{j,k}\,\tfrac{4}{m}\,\tfrac{1}{\pi^2}.
\end{aligned}$$

This leads to the Taylor approximation $\widehat{\text{IOV}} \approx \text{IOV} + \mathbf{D}\,\big(\overline{\boldsymbol{Z}}_t^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\big) + \tfrac{1}{2}\,\big(\overline{\boldsymbol{Z}}_t^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\big)^\top\,\mathbf{H}\,\big(\overline{\boldsymbol{Z}}_t^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\big)$, which is used to conclude the $\widehat{\text{IOV}}$'s asymptotic variance,

$$\begin{aligned}
\tfrac{1}{n}\,\mathbf{D}\,\boldsymbol{\Sigma}\,\mathbf{D}^\top \;&=\; \tfrac{1}{n}\Big(d_{-1}^2\,\sigma_{-1,-1} + 2\sum_{j=0}^{m-1} d_{-1}\,d_j\,\sigma_{-1,j} + \sum_{i,j=0}^{m-1} d_i\,d_j\,\sigma_{i,j}\Big)\\
&\overset{(3.6)}{=}\; \tfrac{1}{n}\Big(d_{-1}^2\,\sigma_{-1,-1} + 2\,d_{-1}\,\big(\sum_{j=0}^{m-1} d_j\,f_j\big)\,\sigma_{-1,-1}\\
&\qquad + \big(\sum_{i=0}^{m-1} d_i\,f_i\big)\,\big(\sum_{j=0}^{m-1} d_j\,f_j\big)\,\sigma_{-1,-1} + \pi^2 \sum_{i,j=0}^{m-1} d_i\,d_j\,\sigma^*_{i,j}\Big)\\
&=\; \tfrac{1}{n}\,\pi^2 \sum_{i,j=0}^{m-1} d_i\,d_j\,\sigma^*_{i,j} \;=\; \tfrac{1}{n}\,\tfrac{16}{m^2} \sum_{i,j=-1}^{m-1} (1 - 2\,f_i)\,(1 - 2\,f_j)\,\sigma^*_{i,j}.
\end{aligned}$$

Analogously, using that $h_{j,k} = 0$ for $j \neq k$, the asymptotic bias becomes

$$\begin{aligned}
&\tfrac{1}{n}\Big(\tfrac{1}{2}\,h_{-1,-1}\,\sigma_{-1,-1} + \sum_{j=0}^{m-1} h_{-1,j}\,\sigma_{-1,j} + \tfrac{1}{2}\sum_{i=0}^{m-1} h_{i,i}\,\sigma_{i,i}\Big)\\
&\overset{(3.6)}{=}\; \tfrac{1}{n}\,\tfrac{4}{m}\,\tfrac{1}{\pi^2}\Big(\big(\sum_{i=0}^{m-1} f_i(1 - 3\,f_i)\big)\,\sigma_{-1,-1} - \big(\sum_{j=0}^{m-1} f_j(1 - 4\,f_j)\big)\,\sigma_{-1,-1}\\
&\qquad - \sum_{i=0}^{m-1}\big(f_i^2\,\sigma_{-1,-1} + \pi^2\,\sigma^*_{i,i}\big)\Big) \;=\; -\tfrac{1}{n}\,\tfrac{4}{m} \sum_{i=0}^{m-1} \sigma^*_{i,i}.
\end{aligned}$$

So the derivations required for Section 3.3 are complete.

S.3 Derivations of Section 3.4

The mean of the $2\,(m+1)$-dimensional binary vectors

$$\begin{aligned}
\boldsymbol{Z}_t^{(h)*} &= \Big(Z_{t,-2}^{(h)*},\quad Z_{t,-1}^{(h)*},\ \overbrace{\ldots, Z_{t,i}^{(h)*}, \ldots}^{i=0,\ldots,m-1},\ \overbrace{\ldots, Z_{t,m+j}^{(h)*}, \ldots}^{j=0,\ldots,m-1}\Big)^\top \\
&:= \Big(O_t\, O_{t-h},\ \underbrace{O_t,\ \ldots, O_t\, Z_{t,i}, \ldots}_{=\,\boldsymbol{Z}_t^*},\ \ldots, O_t\, O_{t-h}\, Z_{t,j}\, Z_{t-h,j}, \ldots\Big)^\top
\end{aligned}$$

defined in (3.11) is equal to

$$E\big[\boldsymbol{Z}_t^{(h)*}\big] \;=\; \Big(\pi(h),\ \pi, \pi\,\boldsymbol{f}^\top,\ \ldots, \pi(h)\, f_{jj}(h), \ldots\Big)^\top \;=:\; \boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}},$$

also recall Section 3.2. Then, like in (3.5), we conclude that

$$\begin{aligned}
&\sqrt{n}\,\Big(\overline{\boldsymbol{Z}_t^{(h)*}} - \boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}}\Big) \;\xrightarrow{\text{d}}\; \mathrm{N}\big(\boldsymbol{0}, \boldsymbol{\Sigma}^{(h)}\big) \quad \text{with } \boldsymbol{\Sigma}^{(h)} = \big(\sigma_{i,j}^{(h)}\big)_{i,j=-2,\ldots,2m-1}, \quad \text{where} \\
&\sigma_{i,j}^{(h)} \;=\; Cov\big[Z_{0,i}^{(h)*}, Z_{0,j}^{(h)*}\big] \;+\; \textstyle\sum_{k=1}^\infty \Big(Cov\big[Z_{0,i}^{(h)*}, Z_{k,j}^{(h)*}\big] + Cov\big[Z_{k,i}^{(h)*}, Z_{0,j}^{(h)*}\big]\Big).
\end{aligned} \tag{S.1}$$

Note that $\sigma_{i,j}^{(h)} = \sigma_{i,j}$ from (3.6) for $i,j = -1, \ldots, m-1$. The remaining covariances involve joint higher-order moments of $(\boldsymbol{Z}_t)$ and (O_t). Let us first compute the covariances referring to the amplitude-modulating process (O_t):

$$\begin{aligned}
\sigma_{-1,-2}^{(h)} &= Cov\big[O_0, O_0 O_{-h}\big] \;+\; \textstyle\sum_{k=1}^\infty \Big(Cov\big[O_0, O_k O_{k-h}\big] + Cov\big[O_k, O_0 O_{-h}\big]\Big) \\
&= (1-\pi)\,\pi(h) \;+\; \textstyle\sum_{k=1}^\infty \Big(E\big[O_0\, O_k O_{k-h}\big] + E\big[O_k\, O_0 O_{-h}\big] - 2\,\pi\,\pi(h)\Big),
\end{aligned}$$

and

$$\begin{aligned}
\sigma_{-2,-2}^{(h)} &= V\big[O_0 O_{-h}\big] \;+\; 2\, \textstyle\sum_{k=1}^\infty Cov\big[O_k O_{k-h}, O_0 O_{-h}\big] \\
&= \pi(h)\,\big(1-\pi(h)\big) \;+\; 2\, \textstyle\sum_{k=1}^\infty \Big(E\big[O_k O_{k-h}\, O_0 O_{-h}\big] - \pi(h)^2\Big).
\end{aligned}$$

For some of the remaining covariances, it is possible to trace them back to $\sigma_{-1,-2}^{(h)}$ and $\sigma_{-2,-2}^{(h)}$:

$$\begin{aligned}
&\sigma_{-2,i}^{(h)} \;=\; Cov\big[O_0 Z_{0,i}, O_0 O_{-h}\big] \;+\; \textstyle\sum_{k=1}^\infty \Big(Cov\big[O_0 Z_{0,i}, O_k O_{k-h}\big] + Cov\big[O_k Z_{k,i}, O_0 O_{-h}\big]\Big) \\
&=\; (1-\pi)\,\pi(h)\, f_i \;+\; \textstyle\sum_{k=1}^\infty \Big(E\big[O_0\, O_k O_{k-h}\big] + E\big[O_k\, O_0 O_{-h}\big] - 2\,\pi\,\pi(h)\Big)\, f_i \;=\; f_i\, \sigma_{-1,-2}^{(h)}; \\
&\sigma_{-2,m+j}^{(h)} \;=\; Cov\big[O_0 O_{-h}, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big] \\
&\quad +\; \textstyle\sum_{k=1}^\infty \Big(Cov\big[O_0 O_{-h}, O_k O_{k-h} Z_{k,j} Z_{k-h,j}\big] + Cov\big[O_k O_{k-h}, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big]\Big) \\
&=\; \pi(h)\,\big(1-\pi(h)\big)\, f_{jj}(h) \;+\; \textstyle\sum_{k=1}^\infty \Big(2\,E\big[O_k O_{k-h}\, O_0 O_{-h}\big] - 2\,\pi(h)^2\Big)\, f_{jj}(h) \;=\; f_{jj}(h)\, \sigma_{-2,-2}^{(h)};
\end{aligned}$$

$$
\begin{aligned}
\sigma^{(h)}_{-1,m+j} \;&=\; Cov\big[O_0, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big] \\
&\quad +\; \textstyle\sum_{k=1}^{\infty} \Big(Cov\big[O_0, O_k O_{k-h} Z_{k,j} Z_{k-h,j}\big] + Cov\big[O_k, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big]\Big) \\
&=\; (1-\pi)\,\pi(h)\, f_{jj}(h) \;+\; \textstyle\sum_{k=1}^{\infty} \Big(E\big[O_0\, O_k O_{k-h}\big] + E\big[O_k\, O_0 O_{-h}\big] - 2\,\pi\,\pi(h)\Big)\, f_{jj}(h) \\
&=\; f_{jj}(h)\, \sigma^{(h)}_{-1,-2}.
\end{aligned}
$$

The expressions for $\sigma^{(h)}_{i,m+j}$ and $\sigma^{(h)}_{m+i,m+j}$, in turn, are generally more complex:

$$
\begin{aligned}
\sigma^{(h)}_{i,m+j} \;&=\; Cov\big[O_0 Z_{0,i}, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big] \\
&\quad +\; \textstyle\sum_{k=1}^{\infty} \Big(Cov\big[O_0 Z_{0,i}, O_k O_{k-h} Z_{k,j} Z_{k-h,j}\big] + Cov\big[O_k Z_{k,i}, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big]\Big) \\
&=\; \pi(h)\, E\big[Z_{0,i}\, Z_{0,j} Z_{-h,j}\big] - \pi\,\pi(h)\, f_i\, f_{jj}(h) \\
&\quad +\; \textstyle\sum_{k=1}^{\infty} \Big(E\big[O_0\, O_k O_{k-h}\big]\, E\big[Z_{0,i}\, Z_{k,j} Z_{k-h,j}\big] \\
&\qquad + E\big[O_k\, O_0 O_{-h}\big]\, E\big[Z_{k,i}\, Z_{0,j} Z_{-h,j}\big] - 2\,\pi\,\pi(h)\, f_i\, f_{jj}(h)\Big); \\
\sigma^{(h)}_{m+i,m+j} \;&=\; Cov\big[O_0 O_{-h} Z_{0,i} Z_{-h,i}, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big] \\
&\quad +\; \textstyle\sum_{k=1}^{\infty} \Big(Cov\big[O_0 O_{-h} Z_{0,i} Z_{-h,i}, O_k O_{k-h} Z_{k,j} Z_{k-h,j}\big] \\
&\qquad + Cov\big[O_k O_{k-h} Z_{k,i} Z_{k-h,i}, O_0 O_{-h} Z_{0,j} Z_{-h,j}\big]\Big) \\
&=\; \pi(h)\, E\big[Z_{0,i} Z_{-h,i}\, Z_{0,j} Z_{-h,j}\big] - \pi(h)^2\, f_{ii}(h)\, f_{jj}(h) \\
&\quad +\; \textstyle\sum_{k=1}^{\infty} \Big(E\big[O_k O_{k-h}\, O_0 O_{-h}\big]\, E\big[Z_{0,i} Z_{-h,i}\, Z_{k,j} Z_{k-h,j}\big] \\
&\qquad + E\big[O_k O_{k-h}\, O_0 O_{-h}\big]\, E\big[Z_{k,i} Z_{k-h,i}\, Z_{0,j} Z_{-h,j}\big] - 2\,\pi(h)^2\, f_{ii}(h)\, f_{jj}(h)\Big).
\end{aligned}
$$

The aim of Section 3.4 is to derive the asymptotic distribution of Cohen's κ under the null hypothesis of (X_t) being i. i. d. Hence, it is relevant to compute the involved higher-order moments of $\boldsymbol{Z}_t$ under this i. i. d.-assumption. Then, see Lemma A.5.1 in Weiß (2020), these moments are

$$
\begin{aligned}
E\big[Z_{0,i} Z_{-h,i}\, Z_{k,j} Z_{k-h,j}\big] - f_{ii}(h)\, f_{jj}(h) \;&=\; \begin{cases} f^2_{\min\{i,j\}} - f_i^2\, f_j^2 & \text{if } k=0,\\ f_i f_j\, f_{\min\{i,j\}} - f_i^2\, f_j^2 & \text{if } k-h=0,\\ 0 & \text{otherwise;}\end{cases} \\
E\big[Z_{k,i}\, Z_{0,j} Z_{-h,j}\big] - f_i\, f_{jj}(h) \;&=\; \begin{cases} f_j\, f_{\min\{i,j\}} - f_i\, f_j^2 & \text{if } k=0,\\ 0 & \text{otherwise;}\end{cases} \\
E\big[Z_{0,i}\, Z_{k,j} Z_{k-h,j}\big] - f_i\, f_{jj}(h) \;&=\; \begin{cases} f_j\, f_{\min\{i,j\}} - f_i\, f_j^2 & \text{if } k, k-h=0,\\ 0 & \text{otherwise.}\end{cases}
\end{aligned}
$$

Hence, under the assumption of (X_t) being i. i. d.,

$$\begin{aligned}
\sigma^{(h)}_{i,m+j} &= \pi(h)\,\big(f_j\,f_{\min\{i,j\}} - f_i\,f_j^2\big) + (1-\pi)\,\pi(h)\,f_i\,f_j^2 + \pi(h)\,\big(f_j\,f_{\min\{i,j\}} - f_i\,f_j^2\big)\\
&\quad + \textstyle\sum_{k=1}^{\infty}\Big(E\big[O_0\,O_kO_{k-h}\big] + E\big[O_k\,O_0O_{-h}\big] - 2\,\pi\,\pi(h)\Big)\,f_i\,f_j^2\\
&= 2\,\pi(h)\,f_j\,\big(f_{\min\{i,j\}} - f_i\,f_j\big) \;+\; f_i\,f_j^2\,\sigma^{(h)}_{-1,-2};\\
\sigma^{(h)}_{m+i,m+j} &= \pi(h)\,\big(f^2_{\min\{i,j\}} - f_i^2\,f_j^2\big) + 2\,E\big[O_hO_0O_{-h}\big]\,f_if_j\,\big(f_{\min\{i,j\}} - f_if_j\big)\\
&\quad + \pi(h)\,\big(1-\pi(h)\big)\,f_i^2\,f_j^2 \;+\; 2\,\textstyle\sum_{k=1}^{\infty}\Big(E\big[O_kO_{k-h}\,O_0O_{-h}\big] - \pi(h)^2\Big)\,f_i^2\,f_j^2\\
&= \pi(h)\,\big(f^2_{\min\{i,j\}} - f_i^2\,f_j^2\big) + 2\,\pi(h,2h)\,f_if_j\,\big(f_{\min\{i,j\}} - f_if_j\big) \;+\; f_i^2\,f_j^2\,\sigma^{(h)}_{-2,-2}.
\end{aligned}$$

Here, we used the notation $\pi(h,2h) = E\big[O_hO_0O_{-h}\big]$.

To obtain the joint asymptotic distribution of $\hat{f}_i^*, \hat{f}_{jj}^*(h)$ according to (3.2), (3.10), we now define the function $\boldsymbol{g} : [0;1]^{2(m+1)} \to [0;1]^{2m}$ by

$$g_j(x_{-2}, x_{-1}, x_0, \ldots, x_{m-1}, x_m, \ldots, x_{2m-1}) \;=\; \begin{cases} x_j/x_{-1} & \text{for } j = 0,\ldots,m-1,\\ x_j/x_{-2} & \text{for } j = m,\ldots,2m-1.\end{cases}$$

Hence,

$$\left.\begin{aligned} g_j\big(\boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}}\big) &= f_j\\ g_{m+j}\big(\boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}}\big) &= f_{jj}(h)\end{aligned}\right\}\quad \text{for } j = 0,\ldots,m-1,$$

and $g_j\Big(\overline{\boldsymbol{Z}_t^{(h)*}}\Big)$ refers to the respective sample counterpart. Like in Section 3.2, the Jacobian of $\boldsymbol{g}$ is sparse; the only non-zero entries are (for $j = 0,\ldots,m-1$)

$$\tfrac{\partial}{\partial x_{-1}}\,g_j \;=\; \tfrac{-x_j}{x_{-1}^2},\qquad \tfrac{\partial}{\partial x_j}\,g_j \;=\; \tfrac{1}{x_{-1}},\qquad \tfrac{\partial}{\partial x_{-2}}\,g_{m+j} \;=\; \tfrac{-x_{m+j}}{x_{-2}^2},\qquad \tfrac{\partial}{\partial x_{m+j}}\,g_{m+j} \;=\; \tfrac{1}{x_{-2}}.$$

Thus, with $\mathbf{D}$ being the Jacobian of $\boldsymbol{g}$ evaluated in $\boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}}$, we have for $j = 0,\ldots,m-1$ that

$$d_{j,-1} \;=\; -\tfrac{f_j}{\pi},\qquad d_{j,j} \;=\; \tfrac{1}{\pi},\qquad d_{m+j,-2} \;=\; \tfrac{-f_{jj}(h)}{\pi(h)},\qquad d_{m+j,m+j} \;=\; \tfrac{1}{\pi(h)}.$$

The Delta method implies that $\sqrt{n}\,\big(\ldots, \hat{f}_i^* - f_i, \ldots, \hat{f}_{jj}^*(h) - f_{jj}(h), \ldots\big)$ is approximately normally distributed with mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}^{(h)*} = \mathbf{D}\,\boldsymbol{\Sigma}^{(h)}\,\mathbf{D}^\top$. Here, we have that $\sigma^{(h)*}_{ij} = \sigma^*_{ij}$ from (3.7) for $i,j = 0,\ldots,m-1$. The remaining entries $\sigma^{(h)*}_{i,m+j}, \sigma^{(h)*}_{m+i,m+j}$ with $i,j = 0,\ldots,m-1$ compute as

$$\begin{aligned}
\sigma^{(h)*}_{i,m+j} &= d_{i,-1}\,d_{m+j,-2}\,\sigma^{(h)}_{-1,-2} \;+\; d_{i,-1}\,d_{m+j,m+j}\,\sigma^{(h)}_{-1,m+j}\\
&\quad + d_{i,i}\,d_{m+j,-2}\,\sigma^{(h)}_{i,-2} \;+\; d_{i,i}\,d_{m+j,m+j}\,\sigma^{(h)}_{i,m+j}\\
&= \tfrac{1}{\pi\,\pi(h)}\,\Big(f_i\,f_{jj}(h)\,\sigma^{(h)}_{-1,-2} - f_i\,\sigma^{(h)}_{-1,m+j} - f_{jj}(h)\,\sigma^{(h)}_{i,-2} + \sigma^{(h)}_{i,m+j}\Big)\\
&= \tfrac{1}{\pi\,\pi(h)}\,\Big(\sigma^{(h)}_{i,m+j} - f_i\,f_{jj}(h)\,\sigma^{(h)}_{-1,-2}\Big),
\end{aligned}$$

and

$$\begin{aligned}
\sigma^{(h)*}_{m+i,m+j} &= d_{m+i,-2}\, d_{m+j,-2}\, \sigma^{(h)}_{-2,-2} \;+\; d_{m+i,-2}\, d_{m+j,m+j}\, \sigma^{(h)}_{-2,m+j} \\
&\quad + d_{m+i,m+i}\, d_{m+j,-2}\, \sigma^{(h)}_{m+i,-2} \;+\; d_{m+i,m+i}\, d_{m+j,m+j}\, \sigma^{(h)}_{m+i,m+j} \\
&= \tfrac{1}{\pi(h)^2} \Big(f_{ii}(h) f_{jj}(h)\, \sigma^{(h)}_{-2,-2} - f_{ii}(h)\, \sigma^{(h)}_{-2,m+j} - f_{jj}(h)\, \sigma^{(h)}_{m+i,-2} + \sigma^{(h)}_{m+i,m+j} \Big) \\
&= \tfrac{1}{\pi(h)^2} \Big(\sigma^{(h)}_{m+i,m+j} - f_{ii}(h) f_{jj}(h)\, \sigma^{(h)}_{-2,-2} \Big).
\end{aligned}$$

If (X_t) is i. i. d., the expressions for $\sigma^{(h)}_{i,m+j}$ and $\sigma^{(h)}_{m+i,m+j}$ further simplify. Together with Example 3, it follows that

$$\begin{aligned}
\sigma^{(h)*}_{i,j} &= \sigma^{*}_{ij} \;=\; \tfrac{1}{\pi} \big(f_{\min\{i,j\}} - f_i f_j \big), \\
\sigma^{(h)*}_{i,m+j} &= \tfrac{2}{\pi}\, f_j \big(f_{\min\{i,j\}} - f_i\, f_j \big) \;=\; 2\, f_j\, \sigma^{(h)*}_{i,j}, \\
\sigma^{(h)*}_{m+i,m+j} &= \tfrac{1}{\pi(h)^2} \big(f_{\min\{i,j\}} - f_i f_j \big) \Big(\pi(h) \big(f_{\min\{i,j\}} + f_i f_j \big) + 2\, \pi(h, 2h)\, f_i f_j \Big).
\end{aligned}$$

So $\sigma^{(h)*}_{m+i,m+j}$ still depends on joint higher-order moments of (O_t). In the special case where also (O_t) is i. i. d., we have $\pi(h) = \pi^2$ and $\pi(h, 2h) = \pi^3$ for $h \geq 1$. So, $\sigma^{(h)*}_{m+i,m+j}$ further simplifies in this case:

$$\sigma^{(h)*}_{m+i,m+j} \;=\; \frac{1}{\pi^2} \big(f_{\min\{i,j\}} - f_i f_j \big) \big(f_{\min\{i,j\}} + (1 + 2\,\pi)\, f_i f_j \big).$$

As the last step, to obtain an approximate distribution for $\hat{\kappa}_{\mathrm{ord}}(h)$ from (3.9) under the null hypothesis of (X_t) being i. i. d., we adapt the derivations in Supplement A.7 of Weiß (2020). Accordingly, we conclude on an approximate normal distribution with variance

$$\begin{aligned}
&\tfrac{1}{n}\, \tfrac{1}{\big(\sum_{k=0}^{m-1} f_k(1-f_k)\big)^2} \sum_{i,j=0}^{m-1} \Big(4\, f_i f_j\, \sigma^{(h)*}_{i,j} \;-\; 4\, f_i\, \sigma^{(h)*}_{i,m+j} \;+\; \sigma^{(h)*}_{m+i,m+j} \Big) \\
&= \tfrac{1}{n}\, \tfrac{1}{\big(\sum_{k=0}^{m-1} f_k(1-f_k)\big)^2}\, \tfrac{1}{\pi(h)} \sum_{i,j=0}^{m-1} \big(f_{\min\{i,j\}} - f_i f_j \big) \Big(f_{\min\{i,j\}} + \big(1 + 2\, \tfrac{\pi(h,2h)}{\pi(h)} - 4\, \tfrac{\pi(h)}{\pi} \big)\, f_i f_j \Big).
\end{aligned}$$

Furthermore, while the true value of $\kappa_{\mathrm{ord}}(h) = 0$ because of the i. i. d.-assumption, the estimator $\hat{\kappa}_{\mathrm{ord}}(h)$ is biased. The bias correction for $n\, \hat{\kappa}_{\mathrm{ord}}(h)$ is computed as

$$\begin{aligned}
&- \tfrac{\sum_{i=0}^{m-1} \sigma^{(h)*}_{i,i}}{\sum_{k=0}^{m-1} f_k(1-f_k)} \;+\; \tfrac{\sum_{i,j=0}^{m-1} (f_i + f_j - 4\, f_i f_j)\, \sigma^{(h)*}_{i,j}}{\big(\sum_{k=0}^{m-1} f_k(1-f_k)\big)^2} \;-\; \tfrac{\sum_{i,j=0}^{m-1} (1 - 2\, f_i)\, \sigma^{(h)*}_{i,m+j}}{\big(\sum_{k=0}^{m-1} f_k(1-f_k)\big)^2} \\
&= \;-\; \tfrac{\sum_{i=0}^{m-1} \sigma^{(h)*}_{i,i}}{\sum_{k=0}^{m-1} f_k(1-f_k)} \;+\; \tfrac{\sum_{i,j=0}^{m-1} (f_i - f_j)\, \sigma^{(h)*}_{i,j}}{\big(\sum_{k=0}^{m-1} f_k(1-f_k)\big)^2} \;=\; -\, \tfrac{1}{\pi} \;+\; 0.
\end{aligned}$$

This completes the proof of Theorem 3.

S.4 Derivations of Section 5 (Sketch)

If (X_t) is a nominal process, with stationary marginal PMF $\boldsymbol{p} = (p_0, \ldots, p_m)^\top \in [0;1]^{m+1}$ and bivariate lag-h probabilities $p_{ij}(h)$, we consider the binarization $(\boldsymbol{Y}_t)$ given by $Y_{t,i} = \mathbb{1}_{\{X_t = s_i\}}$ for $i \in \{0, \ldots, m\}$. Let (O_t) again denote the amplitude-modulating process, which is assumed to be independent of (X_t) (and thus of $(\boldsymbol{Y}_t)$). Then, the amplitude modulation of $(\boldsymbol{Y}_t)$ is given by $(O_t \cdot \boldsymbol{Y}_t)$, and we define the estimator $\hat{\mathfrak{p}} := \frac{1}{n} \sum_{t=1}^n O_t\, \boldsymbol{Y}_t$. With the same arguments as in Section 3.1, it follows that $E\big[\hat{\mathfrak{p}}\big] = \big(\frac{1}{n} \sum_{t=1}^n E[O_t]\big)\, \boldsymbol{p}$. Hence, we estimate $\boldsymbol{p}$ by

$$\hat{\boldsymbol{p}}^* \;:=\; \frac{\frac{1}{n} \sum_{t=1}^n O_t\, \boldsymbol{Y}_t}{\frac{1}{n} \sum_{t=1}^n O_t} \;=:\; \hat{\mathfrak{p}} \,/\, \overline{O},$$

see (5.1). If (O_t) is deterministic, so $\hat{\boldsymbol{p}}^* = \sum_{t=1}^n \frac{o_t}{n_{\text{obs}}}\, \boldsymbol{Y}_t$ with $n_{\text{obs}} = \sum_{s=1}^n o_s$, then $E\big[\hat{\mathfrak{p}}\big] = \overline{o}\, \boldsymbol{p}$, and the cross-covariances of $\hat{\boldsymbol{p}}^*$ follow as

$$\begin{aligned} Cov[\hat{p}_i^*, \hat{p}_j^*] \;&=\; \tfrac{1}{n_{\text{obs}}^2} \textstyle\sum_{s,t=1}^n o_t o_s\, Cov[Y_{t,i}, Y_{s,j}] \\ &=\; \tfrac{1}{n_{\text{obs}}}\, p_j\, (\delta_{i,j} - p_i) \;+\; \tfrac{1}{n_{\text{obs}}^2} \textstyle\sum_{s<t} o_t o_s\, \big(p_{ij}(t-s) + p_{ji}(t-s) - 2\, p_i p_j\big). \end{aligned} \tag{S.2}$$

Here, we used that (Weiß, 2013, Appendix B.1)

$$Cov[Y_{t,i}, Y_{s,j}] \;=\; \begin{cases} p_{ij}(t-s) - p_i p_j & \text{if } t \geq s, \\ p_{ji}(s-t) - p_i p_j & \text{if } s \geq t, \end{cases} \qquad \text{with } p_{ij}(0) = \delta_{i,j}\, p_j.$$

If (O_t) is a stationary stochastic process with $E[O_t] = \pi$, $\gamma_O(h) := Cov[O_h, O_0]$, and $\pi(h) := E[O_h\, O_0] = \gamma_O(h) + \pi^2$, we consider the extended binary vectors

$$\boldsymbol{Y}_t^* = \big(Y_{t,-1}^*, Y_{t,0}^*, \ldots, Y_{t,m}^*\big)^\top \;:=\; \big(O_t,\; O_t\, Y_{t,0}, \ldots, O_t\, Y_{t,m}\big)^\top \;=\; O_t\, \big(1, \boldsymbol{Y}_t^\top\big)^\top,$$

see (5.2). $(\boldsymbol{Y}_t^*)$ is a stationary process, which inherits the mixing assumptions of (X_t) and (O_t) as given in Section 3.2. Its mean equals $\boldsymbol{\mu}_{\boldsymbol{Y}^*} := E\big[\boldsymbol{Y}_t^*\big] = \pi\, \big(1, \boldsymbol{p}^\top\big)^\top$, and we have the CLT

$$\begin{aligned} &\sqrt{n}\, \Big(\overline{\boldsymbol{Y}_t^*} - \boldsymbol{\mu}_{\boldsymbol{Y}^*}\Big) \;\xrightarrow{\text{d}}\; \text{N}(\boldsymbol{0}, \boldsymbol{\Sigma}) \quad \text{with } \boldsymbol{\Sigma} = \big(\sigma_{i,j}\big)_{i,j=-1,\ldots,m}, \quad \text{where} \\ &\sigma_{i,j} \;=\; Cov\big[Y_{0,i}^*, Y_{0,j}^*\big] \;+\; \textstyle\sum_{h=1}^\infty \Big(Cov\big[Y_{0,i}^*, Y_{h,j}^*\big] + Cov\big[Y_{h,i}^*, Y_{0,j}^*\big]\Big). \end{aligned} \tag{S.3}$$

The covariances in (S.3) satisfy

$$Cov\big[Y_{t,i}^*,\; Y_{s,j}^*\big] \;=\; \begin{cases} Cov[O_t,\; O_s] \;= \gamma_O(|t-s|) & \text{if } i = j = -1, \\ Cov[O_t,\; O_s\, Y_{s,j}] & \text{if } i = -1 < j, \\ Cov[O_t\, Y_{t,i},\; O_s\, Y_{s,j}] & \text{if } i,j > -1, \end{cases}$$

so the case $i = j = -1$ is like in Section 3.2. The case $i = -1 < j$ follows with analogous arguments as in Section 3.2, leading to $Cov[O_t,\; O_s\, Y_{s,j}] = \gamma_O(|t-s|)\, p_j$ and $\sigma_{i,-1} = \sigma_{-1,i} = p_i\, \sigma_{-1,-1}$. Finally, for $i,j > -1$, we have

$$Cov[O_t\, Y_{t,i},\; O_s\, Y_{s,j}] \;=\; \pi(|t-s|)\, Cov[Y_{t,i}, Y_{s,j}] + \gamma_O(|t-s|)\, p_i p_j$$

and, thus,

$$\sigma_{i,j} \;=\; p_i p_j\, \sigma_{-1,-1} \;+\; \pi\, p_j\, (\delta_{i,j} - p_i) \;+\; \textstyle\sum_{h=1}^{\infty} \pi(h)\, \big(p_{ij}(h) + p_{ji}(h) - 2\, p_i p_j\big).$$

So altogether, the covariances $\sigma_{i,j}$ in (S.3) compute as

$$\sigma_{i,j} \;=\; \begin{cases} \pi(1-\pi) \;+\; 2\, \sum_{h=1}^{\infty} \gamma_O(h) & \text{if } i = j = -1,\\ p_i\, \sigma_{-1,-1} & \text{if } i > j = -1,\\ p_i p_j\, \sigma_{-1,-1} \;+\; \pi\, p_j\, (\delta_{i,j} - p_i) & \text{if } i, j > -1,\\ \quad +\; \sum_{h=1}^{\infty} \pi(h)\, \big(p_{ij}(h) + p_{ji}(h) - 2\, p_i p_j\big) & \end{cases}$$

which proves (5.3).

S.4.1 Remark It should be noted that some ordinal derivations could be traced back to the nominal derivations by exploring the relation between PMF and CDF. More precisely, defining the triangular "integration matrix" $\mathbf{A} \in \{0,1\}^{m\times(m+1)}$ by

$$\mathbf{A} \;=\; \begin{pmatrix} 1 & 0 & \dots & 0\\ \vdots & \ddots & \ddots & \vdots\\ 1 & \dots & 1 & 0 \end{pmatrix}, \qquad \text{we have} \quad \boldsymbol{f} \;=\; \mathbf{A}\,\boldsymbol{p} \quad \text{as well as} \quad \boldsymbol{Z}_t \;=\; \mathbf{A}\,\boldsymbol{Y}_t,$$

recall (2.1) and (2.2). Extending $\mathbf{A}$ to

$$\mathbf{B} \;=\; \begin{pmatrix} 1 & \mathbf{0}^\top\\ \mathbf{0} & \mathbf{A} \end{pmatrix} \in \{0,1\}^{(m+1)\times(m+2)}, \qquad \text{we also have} \quad \boldsymbol{Z}_t^* \;=\; \mathbf{B}\,\boldsymbol{Y}_t^*,$$

recall (3.4) and (5.2). Therefore, the CLT (S.3) could be used to derive (3.5), and (3.6) could be concluded from (5.3), as follows. We have

$$E\big[\boldsymbol{Z}_t^*\big] \;=\; \mathbf{B}\, E\big[\boldsymbol{Y}_t^*\big] \;=\; \pi\, \mathbf{B}\, \big(1, \boldsymbol{p}^\top\big)^\top \;=\; \pi\, \big(1, \boldsymbol{f}^\top\big)^\top,$$

and

$$Cov\big[\boldsymbol{Z}_t^*, \boldsymbol{Z}_{t-h}^*\big] \;=\; \mathbf{B}\, Cov\big[\boldsymbol{Y}_t^*, \boldsymbol{Y}_{t-h}^*\big]\, \mathbf{B}^\top, \qquad \text{thus} \quad \boldsymbol{\Sigma}_{\boldsymbol{Z}^*} \;=\; \mathbf{B}\, \boldsymbol{\Sigma}_{\boldsymbol{Y}^*}\, \mathbf{B}^\top.$$

The matrix multiplications by $\mathbf{B}$ imply an accumulation of the entries of $\boldsymbol{\Sigma}_{\boldsymbol{Y}^*}$, so (3.6) is derived from (5.3) by doing these summations. But since the derivation of (3.6) in Supplement S.1 is short anyway, and for making the derivations more self-contained, we present the separat derivations here.

The asymptotics of $\hat{\boldsymbol{p}}^*$ are derived nearly identically as in Section 3.2, we just have to increase the dimensionality of $\boldsymbol{g}$ by 1, namely $\boldsymbol{g} : [0;1]^{m+2} \to [0;1]^{m+1}$ with $g_j(x_{-1}, x_0, \dots, x_m) = x_j/x_{-1}$ for $j = 0, \dots, m$. Plugging-in $\boldsymbol{\mu}_{\boldsymbol{Y}^*}$ into the Jacobian of $\boldsymbol{g}$, we get the matrix

$$\mathbf{D} \;=\; \begin{pmatrix} -\frac{p_0}{\pi} & \frac{1}{\pi} & \cdots & 0\\ \vdots & & \ddots & \\ -\frac{p_m}{\pi} & 0 & \cdots & \frac{1}{\pi} \end{pmatrix},$$

and the entries of $\mathbf{\Sigma}^* = \mathbf{D}\,\mathbf{\Sigma}\,\mathbf{D}^\top$ compute as $\sigma^*_{ij} = \frac{1}{\pi^2}\big(\sigma_{i,j} - p_i p_j\,\sigma_{-1,-1}\big)$. So using the Delta method, we conclude that

$$\begin{array}{l}\sqrt{n}\,\Big(\hat{\boldsymbol{p}}^* - \boldsymbol{p}\Big) \xrightarrow{\text{d}} \mathrm{N}\big(\mathbf{0},\ \mathbf{\Sigma}^*\big), \quad \text{with } \mathbf{\Sigma}^* = (\sigma^*_{ij})_{i,j=0,\ldots,m-1}, \quad \text{where} \\ \sigma^*_{ij} \;=\; \frac{1}{\pi}\,p_j\,(\delta_{i,j} - p_i) \;+\; \frac{1}{\pi^2}\sum_{h=1}^\infty \pi(h)\,\big(p_{ij}(h) + p_{ji}(h) - 2\,p_i p_j\big),\end{array}$$

which proves (5.4). Also the bias calculations are nearly identical to those of Section 3.2. Replacing f_j by p_j, we obtain that the bias of $\hat{\boldsymbol{p}}^*$ is of order $o(n^{-1})$.

Result (5.4) can now be applied to derive the asymptotic properties of the sample IQV, recall Section 2 and see the analogous derivations for the sample IOV in Section 3.3. Using the linear Taylor approximation $\widehat{\mathrm{IQV}} \approx \mathrm{IQV} - 2\,\frac{m+1}{m}\sum_{i=0}^m p_i\,(\hat{p}^*_i - p_i)$, we conclude on asymptotic normality with variance

$$V\big[\widehat{\mathrm{IQV}}\big] \;\overset{\text{a}}{=}\; \frac{1}{n}\,\frac{(m+1)^2}{m^2}\,\sum_{i,j=0}^m p_i p_j\,\sigma^*_{ij}.$$

To compute the mean of $\widehat{\mathrm{IQV}}$, we use that $E\big[(\hat{p}^*_i)^2\big] = V\big[\hat{p}^*_i\big] + E\big[\hat{p}^*_i\big]^2$ and conclude like in Section 3.3 that

$$\begin{aligned} E\big[\widehat{\mathrm{IQV}}\big] \;&=\; \tfrac{m+1}{m}\,\Big(1 - \textstyle\sum_{i=0}^m \big(E\big[\hat{p}^*_i\big]^2 + V\big[\hat{p}^*_i\big]\big)\Big) \;\approx\; \mathrm{IQV} \;-\; \tfrac{1}{n}\,\tfrac{m+1}{m}\,\textstyle\sum_{i=0}^m \sigma^*_{ii} \\ &\overset{(5.4)}{=}\; \mathrm{IQV} \;-\; \tfrac{1}{n}\,\tfrac{1}{\pi}\,\mathrm{IQV} \;-\; \tfrac{1}{n}\,\tfrac{2}{\pi^2}\,\tfrac{m+1}{m}\,\textstyle\sum_{h=1}^\infty \pi(h)\,\sum_{i=0}^m \big(p_{ii}(h) - p_i^2\big) \\ &=\; \mathrm{IQV}\,\Big(1 \;-\; \tfrac{1}{n}\,\big(\tfrac{1}{\pi} + \tfrac{2}{\pi^2}\,\textstyle\sum_{h=1}^\infty \pi(h)\,\kappa_{\text{nom}}(h)\big)\Big). \end{aligned}$$

Finally, let us turn to the measurement of nominal serial dependence. In analogy to Section 3.4, we define the bivariate relative frequency $\hat{\mathfrak{p}}_{ij}(h)$ of the amplitude-modulated binarization $(O_t\,\boldsymbol{Y}_t)$ by

$$\hat{\mathfrak{p}}_{ij}(h) \;=\; \frac{1}{n-h}\sum_{t=h+1}^n O_t\,O_{t-h}\,Y_{t,i}\,Y_{t-h,j}.$$

By the independence of (O_t) to (X_t), we get $E\big[\hat{\mathfrak{p}}_{ij}(h)\big] = \big(\frac{1}{n-h}\sum_{t=h+1}^n E[O_t\,O_{t-h}]\big)\,f_{ij}(h)$. This justifies to define the estimator $\hat{p}^*_{ij}(h)$ as in (5.7), i. e., as

$$\hat{p}^*_{ij}(h) \;=\; \frac{\frac{1}{n-h}\sum_{t=h+1}^n O_t\,O_{t-h}\,Y_{t,i}\,Y_{t-h,j}}{\frac{1}{n-h}\sum_{t=h+1}^n O_t\,O_{t-h}} \;=:\; \hat{\mathfrak{p}}_{ij}(h)\,\big/\,\overline{O_t O_{t-h}}$$

in analogy to (5.1). If (O_t) is again assumed to be stationary, then $E\big[\hat{\mathfrak{p}}_{ij}(h)\big] = \pi(h)\,p_{ij}(h)$ such that $\hat{p}^*_{ij}(h)$ from (5.7) is generally biased. To derive the asymptotics of the $\hat{p}^*_{ij}(h)$, we define the $2\,(m+2)$-dimensional binary vectors

$$\begin{aligned} \boldsymbol{Y}^{(h)*}_t \;&=\; \big(Y^{(h)*}_{t,-2},\quad Y^{(h)*}_{t,-1},\ \overbrace{\ldots, Y^{(h)*}_{t,i}, \ldots}^{i=0,\ldots,m},\ \overbrace{\ldots, Y^{(h)*}_{t,m+1+j}, \ldots}^{j=0,\ldots,m}\big)^\top \\ :&=\; \big(O_t\,O_{t-h},\ \underbrace{O_t,\ \ldots, O_t\,Y_{t,i}, \ldots}_{=\,\boldsymbol{Y}^*_t},\ \ldots, O_t\,O_{t-h}\,Y_{t,j}\,Y_{t-h,j}, \ldots\big)^\top, \end{aligned} \qquad \text{(S.4)}$$

where the components numbered by $-1, 0, \ldots, m$ agree with those of $\boldsymbol{Y}_t^*$ in (5.2), and where the additional components at positions -2 and $m+1, \ldots, 2m+1$ refer to the relevant interactions at times t and $t-h$. The mean of $\boldsymbol{Y}_t^{(h)*}$ is equal to

$$E\big[\boldsymbol{Y}_t^{(h)*}\big] \;=\; \Big(\pi(h),\ \pi, \pi\,\boldsymbol{p}^\top,\ \ldots, \pi(h)\,p_{jj}(h), \ldots\Big)^\top \;=: \boldsymbol{\mu}_{\boldsymbol{Y}^{(h)*}},$$

and we conclude that

$$\begin{aligned} &\sqrt{n}\left(\overline{\boldsymbol{Y}_t^{(h)*}} - \boldsymbol{\mu}_{\boldsymbol{Y}^{(h)*}}\right) \;\xrightarrow{\text{d}}\; \mathrm{N}\big(\boldsymbol{0}, \boldsymbol{\Sigma}^{(h)}\big) \quad \text{with } \boldsymbol{\Sigma}^{(h)} = \big(\sigma_{i,j}^{(h)}\big)_{i,j=-2,\ldots,2m+1}, \quad \text{where} \\ &\sigma_{i,j}^{(h)} \;=\; Cov\big[Y_{0,i}^{(h)*}, Y_{0,j}^{(h)*}\big] \;+\; \textstyle\sum_{k=1}^\infty \Big(Cov\big[Y_{0,i}^{(h)*}, Y_{k,j}^{(h)*}\big] + Cov\big[Y_{k,i}^{(h)*}, Y_{0,j}^{(h)*}\big]\Big). \end{aligned} \tag{S.5}$$

We have $\sigma_{i,j}^{(h)} = \sigma_{i,j}$ from (5.3) for $i,j = -1, \ldots, m$, and $\sigma_{-1,-2}^{(h)}, \sigma_{-2,-2}^{(h)}$ are the same as in Section 3.4, i. e.,

$$\begin{aligned} \sigma_{-1,-2}^{(h)} &= (1-\pi)\,\pi(h) \;+\; \textstyle\sum_{k=1}^\infty \Big(E\big[O_0\,O_k O_{k-h}\big] + E\big[O_k\,O_0 O_{-h}\big] - 2\,\pi\,\pi(h)\Big), \\ \sigma_{-2,-2}^{(h)} &= \pi(h)\,\big(1-\pi(h)\big) \;+\; 2\,\textstyle\sum_{k=1}^\infty \Big(E\big[O_k O_{k-h}\,O_0 O_{-h}\big] - \pi(h)^2\Big). \end{aligned}$$

Also the derivations of the remaining covariances in Section 3.4 require only few adaptions. We get

$$\sigma_{-2,i}^{(h)} \;=\; p_i\,\sigma_{-1,-2}^{(h)}, \qquad \sigma_{-2,m+1+j}^{(h)} \;=\; p_{jj}(h)\,\sigma_{-2,-2}^{(h)}, \qquad \sigma_{-1,m+1+j}^{(h)} \;=\; p_{jj}(h)\,\sigma_{-1,-2}^{(h)}$$

for $i,j = 0, \ldots, m$, and

$$\begin{aligned} \sigma_{i,m+1+j}^{(h)} &= \pi(h)\,E\big[Y_{0,i}\,Y_{0,j}Y_{-h,j}\big] - \pi\,\pi(h)\,p_i\,p_{jj}(h) \\ &\quad+\; \textstyle\sum_{k=1}^\infty \Big(E\big[O_0\,O_kO_{k-h}\big]\,E\big[Y_{0,i}\,Y_{k,j}Y_{k-h,j}\big] \\ &\qquad +E\big[O_k\,O_0O_{-h}\big]\,E\big[Y_{k,i}\,Y_{0,j}Y_{-h,j}\big] - 2\,\pi\,\pi(h)\,p_i\,p_{jj}(h)\Big), \\ \sigma_{m+1+i,m+1+j}^{(h)} &= \pi(h)\,E\big[Y_{0,i}Y_{-h,i}\,Y_{0,j}Y_{-h,j}\big] - \pi(h)^2\,p_{ii}(h)\,p_{jj}(h) \\ &\quad+\; \textstyle\sum_{k=1}^\infty \Big(E\big[O_kO_{k-h}\,O_0O_{-h}\big]\,E\big[Y_{0,i}Y_{-h,i}\,Y_{k,j}Y_{k-h,j}\big] \\ &\qquad +E\big[O_kO_{k-h}\,O_0O_{-h}\big]\,E\big[Y_{k,i}Y_{k-h,i}\,Y_{0,j}Y_{-h,j}\big] - 2\,\pi(h)^2\,p_{ii}(h)\,p_{jj}(h)\Big). \end{aligned}$$

Under the additional assumption of (X_t) being i. i. d. (null hypothesis), we have $p_{ii}(h) = p_i^2$ and $p_{jj}(h) = p_j^2$ for $i,j = 0, \ldots, m$, and Appendix D.1 in Weiß (2013) provides

$$\begin{aligned} E\big[Y_{0,i}Y_{-h,i}\,Y_{k,j}Y_{k-h,j}\big] - p_{ii}(h)\,p_{jj}(h) &= \begin{cases} (\delta_{i,j} - p_i^2)\,p_j^2 & \text{if } k = 0, \\ (\delta_{i,j} - p_i)\,p_i\,p_j^2 & \text{if } k-h = 0, \\ 0 & \text{otherwise;} \end{cases} \\ E\big[Y_{k,i}\,Y_{0,j}Y_{-h,j}\big] - p_i\,p_{jj}(h) &= \begin{cases} (\delta_{i,j} - p_i)\,p_j^2 & \text{if } k = 0, \\ 0 & \text{otherwise;} \end{cases} \\ E\big[Y_{0,i}\,Y_{k,j}Y_{k-h,j}\big] - p_i\,p_{jj}(h) &= \begin{cases} (\delta_{i,j} - p_i)\,p_j^2 & \text{if } k, k-h = 0, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

So we conclude that

$$\begin{aligned} \sigma^{(h)}_{i,m+1+j} &= 2\,\pi(h)\,(\delta_{i,j}-p_i)\,p_j^2 \;+\; p_i\,p_j^2\,\sigma^{(h)}_{-1,-2}, \\ \sigma^{(h)}_{m+1+i,m+1+j} &= \pi(h)\,(\delta_{i,j}-p_i^2)\,p_j^2 + 2\,\pi(h,2h)\,(\delta_{i,j}-p_i)\,p_i\,p_j^2 \;+\; p_i^2\,p_j^2\,\sigma^{(h)}_{-2,-2}, \end{aligned} \tag{S.6}$$

where again $\pi(h,2h) = E\big[O_h O_0 O_{-h}\big]$.

To obtain the joint asymptotic distribution of $\hat{p}_i^*, \hat{p}_{jj}^*(h)$ according to (5.1), (5.7), we now define the function $\boldsymbol{g} : [0;1]^{2(m+2)} \to [0;1]^{2(m+1)}$ by

$$g_j(x_{-2},x_{-1},x_0,\ldots,x_m,x_{m+1},\ldots,x_{2m+1}) \;=\; \begin{cases} x_j/x_{-1} & \text{for } j=0,\ldots,m, \\ x_j/x_{-2} & \text{for } j=m+1,\ldots,2m+1. \end{cases}$$

Hence,

$$\left.\begin{aligned} g_j\big(\boldsymbol{\mu}_{\boldsymbol{Y}^{(h)*}}\big) &= p_j \\ g_{m+1+j}\big(\boldsymbol{\mu}_{\boldsymbol{Y}^{(h)*}}\big) &= p_{jj}(h) \end{aligned}\right\} \quad \text{for } j=0,\ldots,m,$$

and $g_j\Big(\overline{\boldsymbol{Y}_t^{(h)*}}\Big)$ refers to the respective sample counterpart. Like in Section 3.4, the Jacobian of $\boldsymbol{g}$ is sparse; the only non-zero entries are (for $j=0,\ldots,m$)

$$\tfrac{\partial}{\partial x_{-1}}\,g_j \;=\; \tfrac{-x_j}{x_{-1}^2}, \quad \tfrac{\partial}{\partial x_j}\,g_j \;=\; \tfrac{1}{x_{-1}}, \quad \tfrac{\partial}{\partial x_{-2}}\,g_{m+1+j} \;=\; \tfrac{-x_{m+1+j}}{x_{-2}^2}, \quad \tfrac{\partial}{\partial x_{m+1+j}}\,g_{m+1+j} \;=\; \tfrac{1}{x_{-2}}.$$

Thus, with $\mathbf{D}$ being the Jacobian of $\boldsymbol{g}$ evaluated in $\boldsymbol{\mu}_{\boldsymbol{Y}^{(h)*}}$, we have for $j=0,\ldots,m$ that

$$d_{j,-1} \;=\; -\tfrac{p_j}{\pi}, \quad d_{j,j} \;=\; \tfrac{1}{\pi}, \quad d_{m+1+j,-2} \;=\; \tfrac{-p_{jj}(h)}{\pi(h)}, \quad d_{m+1+j,m+1+j} \;=\; \tfrac{1}{\pi(h)}.$$

The Delta method implies that $\sqrt{n}\,\big(\ldots,\hat{p}_i^*-p_i,\ldots,\hat{p}_{jj}^*(h)-p_{jj}(h),\ldots\big)$ is approximately normally distributed with mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}^{(h)*} = \mathbf{D}\,\boldsymbol{\Sigma}^{(h)}\,\mathbf{D}^\top$. Here, we have that $\sigma_{ij}^{(h)*} = \sigma_{ij}^*$ from (5.4) for $i,j=0,\ldots,m$. The remaining entries $\sigma^{(h)*}_{i,m+1+j}, \sigma^{(h)*}_{m+1+i,m+1+j}$ with $i,j=0,\ldots,m$ are computed in the same way as in Section 3.4, leading to

$$\begin{aligned} \sigma^{(h)*}_{i,m+1+j} &= \tfrac{1}{\pi\,\pi(h)}\,\Big(\sigma^{(h)}_{i,m+1+j} - p_i\,p_{jj}(h)\,\sigma^{(h)}_{-1,-2}\Big), \\ \sigma^{(h)*}_{m+1+i,m+1+j} &= \tfrac{1}{\pi(h)^2}\,\Big(\sigma^{(h)}_{m+1+i,m+1+j} - p_{ii}(h)p_{jj}(h)\,\sigma^{(h)}_{-2,-2}\Big). \end{aligned}$$

If (X_t) is i. i. d., the expressions for $\sigma^{(h)*}_{i,j}$, $\sigma^{(h)}_{i,m+j}$, and $\sigma^{(h)}_{m+i,m+j}$ further simplify. Using (5.4) and (S.6), we get

$$\begin{aligned} \sigma^{(h)*}_{i,j} &= \sigma^*_{ij} \;=\; \tfrac{1}{\pi}\,p_j\,(\delta_{i,j}-p_i), \\ \sigma^{(h)*}_{i,m+1+j} &= \tfrac{2}{\pi}\,(\delta_{i,j}-p_i)\,p_j^2 \;=\; 2\,p_j\,\sigma^{(h)*}_{i,j}, \\ \sigma^{(h)*}_{m+1+i,m+1+j} &= \tfrac{1}{\pi(h)^2}\,(\delta_{i,j}-p_i)\,p_j^2\,\Big(\pi(h)\,(\delta_{i,j}+p_i) + 2\,\pi(h,2h)\,p_i\Big). \end{aligned} \tag{S.7}$$

As the last step, we derive an approximate distribution for the nominal Cohen's κ from (5.5),

$$\hat\kappa_{\mathrm{nom}}(h) \;=\; \frac{\sum_{j=0}^m \big(\hat p_{jj}^*(h) - (\hat p_j^*)^2\big)}{1-\sum_{i=0}^m (\hat p_i^*)^2} \qquad \text{for } h\in\mathbb{N},$$

under the null hypothesis of (X_t) being i. i. d. For this purpose, we adapt the derivations in Appendix C.1 of Weiß (2013) and Appendix B.3 of Weiß (2019). Define the function

$$g(x_0,\ldots,x_m,x_{m+1},\ldots,x_{2m+1}) \;:=\; 1-\frac{1-\sum_{j=0}^m x_{m+1+j}}{1-\sum_{i=0}^m x_i^2},$$

such that $g(\ldots,\hat p_i^*,\ldots,\hat p_{jj}^*(h),\ldots)=\hat\kappa_{\mathrm{nom}}(h)$. Then, for $k,l=0,\ldots,m$, we have

$$\tfrac{\partial}{\partial x_k}\, g(\boldsymbol{x}) \;=\; -2x_k\,\frac{1-\sum_{j=0}^m x_{m+1+j}}{(1-\sum_{i=0}^m x_i^2)^2} \;=\; \frac{-2x_k\,\big(1-g(\boldsymbol{x})\big)}{1-\sum_{i=0}^m x_i^2}, \qquad \tfrac{\partial}{\partial x_{m+1+k}}\, g(\boldsymbol{x}) \;=\; \frac{1}{1-\sum_{i=0}^m x_i^2},$$

and

$$\begin{aligned}
\tfrac{\partial^2}{\partial x_k\,\partial x_l}\, g(\boldsymbol{x}) \;&=\; \frac{2x_k\,\frac{\partial}{\partial x_l}\, g(\boldsymbol{x})}{1-\sum_{i=0}^m x_i^2} \;-\; \big(1-g(\boldsymbol{x})\big)\,\tfrac{\partial}{\partial x_l}\,\frac{2x_k}{1-\sum_{i=0}^m x_i^2}\\
&=\; \frac{-4x_kx_l\,\big(1-g(\boldsymbol{x})\big)}{(1-\sum_{i=0}^m x_i^2)^2} \;-\; \big(1-g(\boldsymbol{x})\big)\Big(\frac{2\delta_{k,l}}{1-\sum_{i=0}^m x_i^2}+\frac{4x_kx_l}{(1-\sum_{i=0}^m x_i^2)^2}\Big)\\
&=\; -\big(1-g(\boldsymbol{x})\big)\Big(\frac{2\delta_{k,l}}{1-\sum_{i=0}^m x_i^2}+\frac{8x_kx_l}{(1-\sum_{i=0}^m x_i^2)^2}\Big),\\
\tfrac{\partial^2}{\partial x_{m+1+k}\,\partial x_l}\, g(\boldsymbol{x}) \;&=\; \frac{2x_l}{(1-\sum_{i=0}^m x_i^2)^2}, \qquad \tfrac{\partial^2}{\partial x_{m+1+k}\,\partial x_{m+1+l}}\, g(\boldsymbol{x}) \;=\; 0.
\end{aligned}$$

Evaluated in $(\ldots,p_i,\ldots,p_{jj}(h),\ldots)$, we get

$$d_k \;:=\; \frac{-2\,p_k\,\big(1-\kappa_{\mathrm{nom}}(h)\big)}{1-\sum_{i=0}^m p_i^2}, \qquad d_{m+1+k} \;:=\; \frac{1}{1-\sum_{i=0}^m p_i^2},$$

and

$$\begin{aligned}
h_{k,l} \;&:=\; -2\,\big(1-\kappa_{\mathrm{nom}}(h)\big)\Big(\frac{\delta_{k,l}}{1-\sum_{i=0}^m p_i^2}+\frac{4\,p_kp_l}{(1-\sum_{i=0}^m p_i^2)^2}\Big),\\
h_{m+1+k,l} \;&:=\; \frac{2\,p_l}{(1-\sum_{i=0}^m p_i^2)^2}, \qquad h_{m+1+k,m+1+l} \;=\; 0.
\end{aligned}$$

Since (X_t) is i. i. d., we have $\kappa_{\mathrm{nom}}(h)=0$, which leads to further simplifications.
Using the Delta method, we conclude that $\sqrt{n}\,\hat\kappa_{\mathrm{nom}}(h)$ is asymptotically normally distributed with mean 0 and variance

$$\begin{aligned}
&\textstyle\sum_{i,j=0}^m d_id_j\,\sigma_{i,j}^{(h)*} \;+\; 2\sum_{i,j=0}^m d_id_{m+1+j}\,\sigma_{i,m+1+j}^{(h)*} \;+\; \sum_{i,j=0}^m d_{m+1+i}d_{m+1+j}\,\sigma_{m+1+i,m+1+j}^{(h)*}\\
&\quad=\; \tfrac{1}{(1-\sum_{i=0}^m p_i^2)^2}\Big(4\sum_{i,j=0}^m p_ip_j\,\sigma_{i,j}^{(h)*} \;-\; 4\sum_{i,j=0}^m p_i\,\sigma_{i,m+1+j}^{(h)*} \;+\; \sum_{i,j=0}^m \sigma_{m+1+i,m+1+j}^{(h)*}\Big)\\
&\quad\overset{(S.7)}{=}\; \tfrac{1}{(1-\sum_{i=0}^m p_i^2)^2}\,\tfrac{1}{\pi(h)}\,\sum_{i,j=0}^m p_j^2\,(\delta_{i,j}-p_i)\,\big(\delta_{i,j}+p_i\,(1+2\,\tfrac{\pi(h,2h)}{\pi(h)}-4\,\tfrac{\pi(h)}{\pi})\big).
\end{aligned}$$

A bias correction for $n\,\hat{\kappa}_{\mathrm{nom}}(h)$ is derived based on a second-order Taylor approximation, leading to

$$\begin{aligned}
&\tfrac{1}{2}\ \textstyle\sum_{i,j=0}^{m} h_{i,j}\,\sigma_{i,j}^{(h)*}\ +\ \sum_{i,j=0}^{m} h_{i,m+1+j}\,\sigma_{i,m+1+j}^{(h)*}\\
&\quad=\ \textstyle\sum_{i,j=0}^{m} \frac{2\,p_i\,\sigma_{i,m+1+j}^{(h)*}}{(1-\sum_{r=0}^{m} p_r^2)^2}\ -\ \sum_{i,j=0}^{m} \frac{4\,p_i p_j\,\sigma_{i,j}^{(h)*}}{(1-\sum_{r=0}^{m} p_r^2)^2}\ -\ \sum_{i=0}^{m} \frac{\sigma_{i,i}^{(h)*}}{1-\sum_{r=0}^{m} p_r^2}\\
&\quad\overset{(\mathrm{S}.7)}{=}\ 0\ -\ \tfrac{1}{\pi}\,\frac{\sum_{i=0}^{m} p_i\,(1-p_i)}{1-\sum_{r=0}^{m} p_r^2}\ =\ -\tfrac{1}{\pi}.
\end{aligned}$$